

* In closed-loop settings, the measurements are dependent on the MHE estimates!

$$\hat{x}_k, \hat{d}_k \rightarrow u(\hat{x}_k, \hat{d}_k, x^{ref}) \rightarrow x_k \rightarrow y_k = h(x_k) + v \rightarrow \text{MHE update}$$

Added into my PhD Thesis.

30 - Oct. 2023.

Intuitive Explanation of Online Learning NeuroMHE.

~~Supervised~~ Supervised learning of NeuroMHE is intuitive ~~and~~, straightforward, and effective, as it directly measures the deviation of the MHE estimates away from their ground truth values, which is the core target of state estimation.

$$\hat{x} \xrightarrow{\text{MHE}} x \text{ (ground truth)}$$

Now a question arises, why does RL of NeuroMHE from the tracking errors work, at least in the simulations using the synthetic dataset? Minimization of the error between the MHE estimates and the corresponding reference states is seemingly counter-intuitive for state estimation.

$$\hat{x} \xrightarrow[\substack{R \uparrow \quad Q \downarrow}]{\text{MHE}} x \xrightarrow[\substack{d \rightarrow d}]{\text{Control}} x^{ref} \text{ (reference state)}$$

Below is an explanation of why such a tracking error-based ~~learning~~ learning works. (Note that it works only under closed-loop robust control). At the core of the explanation is to show that minimizing the tracking error functions in the same manner as minimizing the estimation error does! Let's show it by contradiction.

- Our expectation for learning: ~~an~~ Effective learning should lead to ^{an} increasing the weights related to the measurement error while reducing the weights related to the process noises. $R \uparrow \quad Q \downarrow$ (so, we are allowed to change the process noises for tracking the relatively accurate measurement. In this way, the estimated disturbances should converge toward their ground truth values.)

- Opposing results: After training, the weights related to the measurement error decreases but there is a significant increase in the weights for the process noises. $R \downarrow \quad Q \uparrow$ So, we have very limited confidence in the measurement and want to suppress the process noises significantly. (The optimal disturbances will (and more likely to) be zeros.)

We are going to show that the fulfillment of the control objective and the opposing results cannot co-exist! (Note that the feedback controller takes as input the current estimate, including the disturbance estimate, from NeuroMHE)

$$\min_{X, W} J = \frac{1}{2} \|x_0 - \hat{x}_0\|_P^2 + \sum_{k=0}^N \frac{1}{2} \|y_k - h(x_k)\|_R^2 + \sum_{k=0}^{N-1} \frac{1}{2} \|\hat{w}_k\|_Q^2$$

* Initial guess of $\hat{d}_0$ is zero!

$$\text{s.t. } \begin{cases} \hat{x}_{k+1} = f(\hat{x}_k) + \Delta t \cdot (f(\hat{x}_k) + [-f(\hat{x}_k) + \hat{x}_k^{ref} - \hat{d}_k] + \hat{d}_k) \\ \hat{d}_{k+1} = \hat{d}_k + \Delta t \cdot \hat{w}_k \end{cases}$$

$u :=$ numerical value (robust feedback controller taking input for MHE the disturbance estimate as the feedforward term for disturbance compensation)

$$x := [x^0, d]$$

In the opposing results, $R \downarrow \quad Q \uparrow$ result in nearly zero estimate of the disturbance $\hat{d}_k \approx 0$. If R is very small, the resulting MHE and the controller are decoupled from the robotic system, as the MHE estimator does not need real measurement sampled from the real system! This means that the state estimates from MHE need not be related to the measurement / states of the real system, much less the reference states. This is because, in theory, there is no any forms of constraint on the system states in the cost.

Satisfying the dynamics equality constraint does not ensure that the state estimates can track the reference states! $\rightarrow$ The opposing results cannot occur!

When $\hat{d} \rightarrow d$, then $x \rightarrow x^{ref}$, leading to $y \rightarrow y^{ref}$, and this return encourages $\hat{x}$ to become x^{ref}
 else
 When $\hat{d} \nrightarrow d$, then $x \nrightarrow x^{ref}$, leading to $y \nrightarrow y^{ref}$, so $\hat{x}^{MHE}$ deviates from x^{ref} .
 In the MHE cost, $\|y_k - h(x_k)\|$ is a cost that can guide x_k towards x^{ref} through $y \rightarrow y^{ref}$.
 30-Dec-2023.

continued discussion of the tracking error-based RL of NeuroMHE.

2-May-2026 min Loss ($\hat{x}^q(0), x^{q,ref}$)
 For MHE KKT system: $F(z, \lambda; \theta, Y, U) = 0$.
 then exact closed-loop differentiation gives:

$$F_{z,\lambda} \frac{d(z, \lambda)}{d\theta} = -(F_\theta + F_Y \frac{dY}{d\theta} + F_U \frac{dU}{d\theta})$$

 St. $\hat{x}(0), \hat{d}(0) = MHE(y, u, \theta) \rightarrow$ MHE is coupled with the true system by the measurement y sampled from the system.
 $u = u(\hat{x}^q, \hat{d}, x^{q,ref})$
 $y = Model(x, u) \rightarrow$

If $\hat{x}^q$ deviates from y significantly as a result of a very small R , penalizing the measurement error, MHE is then decoupled from the robot system and the stability of the MHE estimates (for the freezer is at risk, data MHE layer)
 $\hat{d} \rightarrow d \iff \hat{x}^q \rightarrow x^{q,ref}$
 We have shown in the left page that $\hat{x}^q \rightarrow x^{q,ref} \Rightarrow \hat{d} \nrightarrow d$ is impossible.
 $\hat{x}^q \rightarrow x^{q,ref}$ is possible if and only if $\hat{d} \rightarrow d$.

So, in other words, (intuitively) $\hat{d} \rightarrow d$ is a major (almost the sole) factor that contributes to the improved robust tracking performance. Minimizing this loss can encourage pushing disturbance the MHE estimate $\hat{d}$ toward the ground truth.
 2-May-2026. For the exact closed-loop objective, we need to consider the partial derivative of the measurement y and the control u w.r.t. θ in MHE!

7-May-2023 Proposed Timeline For Future Research:

Discussions on Whether We need to differentiate external signals?

For our optimization problem: $z^* = \underset{z}{\operatorname{argmin}} J(z, \lambda; \theta, s)$ Its KKT conditions: $F(z, \lambda; \theta, s) = 0$.
 where z is the optimization variable, λ is the Lagrangian dual variable, θ is the learnable parameter, and s is an external input signal / data. We differentiate s only when s itself depends on θ in the computational graph of the final / upper-level loss.

* The exact, complete gradient of z^* w.r.t. θ using chain rule:

$$F_{z,\lambda} \cdot \frac{d(z, \lambda)}{d\theta} = -(F_\theta + F_s \cdot \frac{ds}{d\theta}) \quad * \quad F_\theta := \frac{\partial F}{\partial \theta}, F_s := \frac{\partial F}{\partial s}, F_z := \frac{\partial F}{\partial z}$$

- When we omit $\frac{ds}{d\theta}$?
 When the optimization is treated as a standalone layer with the external input data s fixed e.g., open-loop NeuroMHE trained using the ground truth disturbance data: $\frac{ds}{d\theta} = 0$.
- When we need to consider $\frac{ds}{d\theta}$ for exact gradient computation?
 When s is generated by another part of the closed-loop (or iterative) computational graph and thus depends on θ ! in distributed optimization

For Neuro MHE in closed-loop: $\theta \rightarrow \hat{x} \rightarrow u \rightarrow x \rightarrow y \rightarrow \hat{x}^{ref}$
 so for the full closed-loop exact policy-gradient view, the differential KKT condition should contain terms like:

$$F_{z,\lambda} \frac{d(z, \lambda)}{d\theta} = -(F_\theta + F_Y \frac{dY}{d\theta} + F_U \frac{dU}{d\theta})$$

 Differentiate an external input to an optimization problem only when that input is itself generated by upstream parameter-dependent computation in the final end-to-end learning objective!